

INSTACART CUSTOMER PURCHASE BEHAVIOUR ANALYSIS

Project Name : Instacart Customer Purchase Behaviour

Dataset : Instacart Market Basket Analysis

1.Import Dataset:

```
1 import pandas as pd
2 import numpy as np
3 import matplotlib.pyplot as plt
4 import seaborn as sns
5 #####1. Import Dataset
6 orders = pd.read_csv("orders.csv")
7 products = pd.read_csv("products.csv")
8 order_products = pd.read_csv("order_products__prior.csv")
9 departments = pd.read_csv("departments.csv")
```

Data :

```
PS D:\DATA SCIENCE AND ML> cd "DS_PROJECT(Instacart_Project)"
PS D:\DATA SCIENCE AND ML\DS_PROJECT(Instacart_Project)> python instacart_analysis.py
First 5 Rows
  order_id  user_id  eval_set  order_number  order_dow  order_hour_of_day  days_since_prior_order
0    2539329      1    prior            1           2             8                NaN
1    2398795      1    prior            2           3             7                15.0
2    473747      1    prior            3           3            12                21.0
3    2254736      1    prior            4           4             7                29.0
4    431534      1    prior            5           4            15                28.0
```

2. Basic Data:

----- BASIC DATA -----

Total Customers
206209

Total Columns
7

Missing Values

order_id	0
user_id	0
eval_set	0
order_number	0
order_dow	0
order_hour_of_day	0
days_since_prior_order	206209
dtype: int64	

3. Analysis

```
----- ANALYSIS -----  
  
Total Orders  
3421083  
  
Total Products  
49688  
  
Total Customers  
206209  
  
Average Orders Per Customer  
16.59
```

4. Order Comparison

```
----- ORDER COMPARISON -----  
  
Low Reorder Orders      1078596  
High Reorder Orders     2136278
```

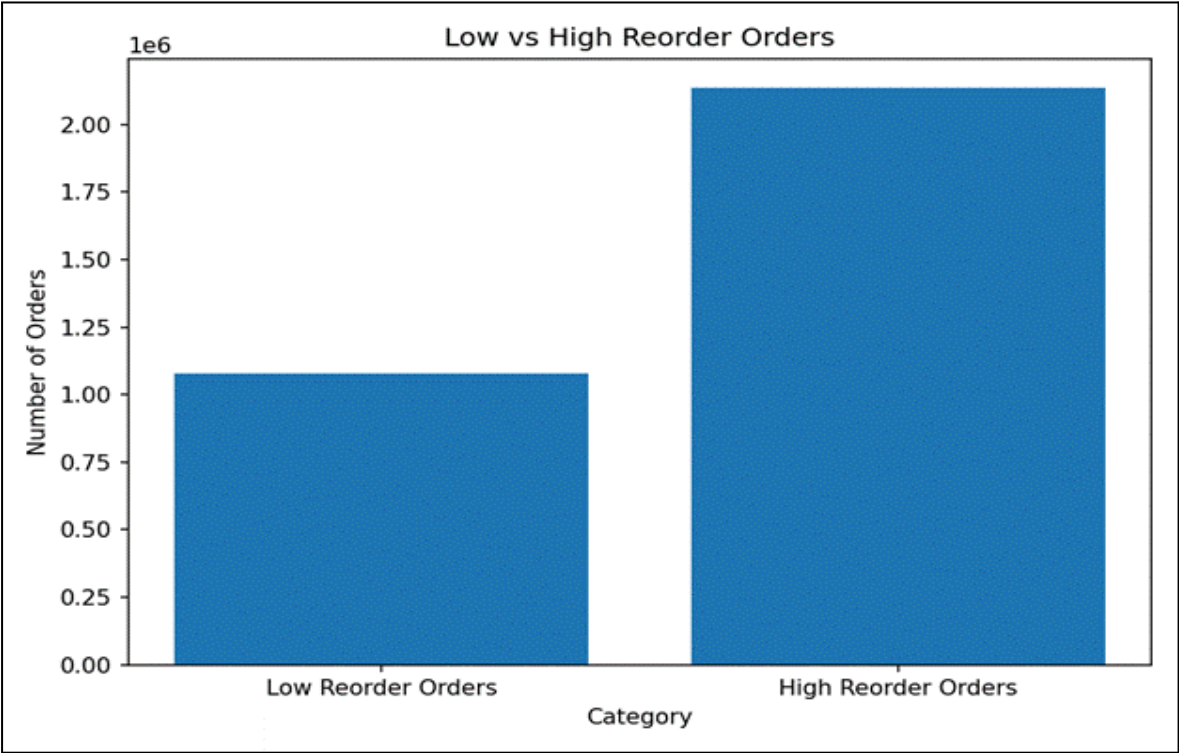


Figure 1 : Low vs High Reorder Orders

5. Product Hotspot:

----- TOP 10 PRODUCTS -----

```
product_name  
Banana 472565  
Bag of Organic Bananas 379450  
Organic Strawberries 264683  
Organic Baby Spinach 241921  
Organic Hass Avocado 213584  
Organic Avocado 176815  
Large Lemon 152657  
Strawberries 142951  
Limes 140627  
Organic Whole Milk 137905  
Name: count, dtype: int64
```

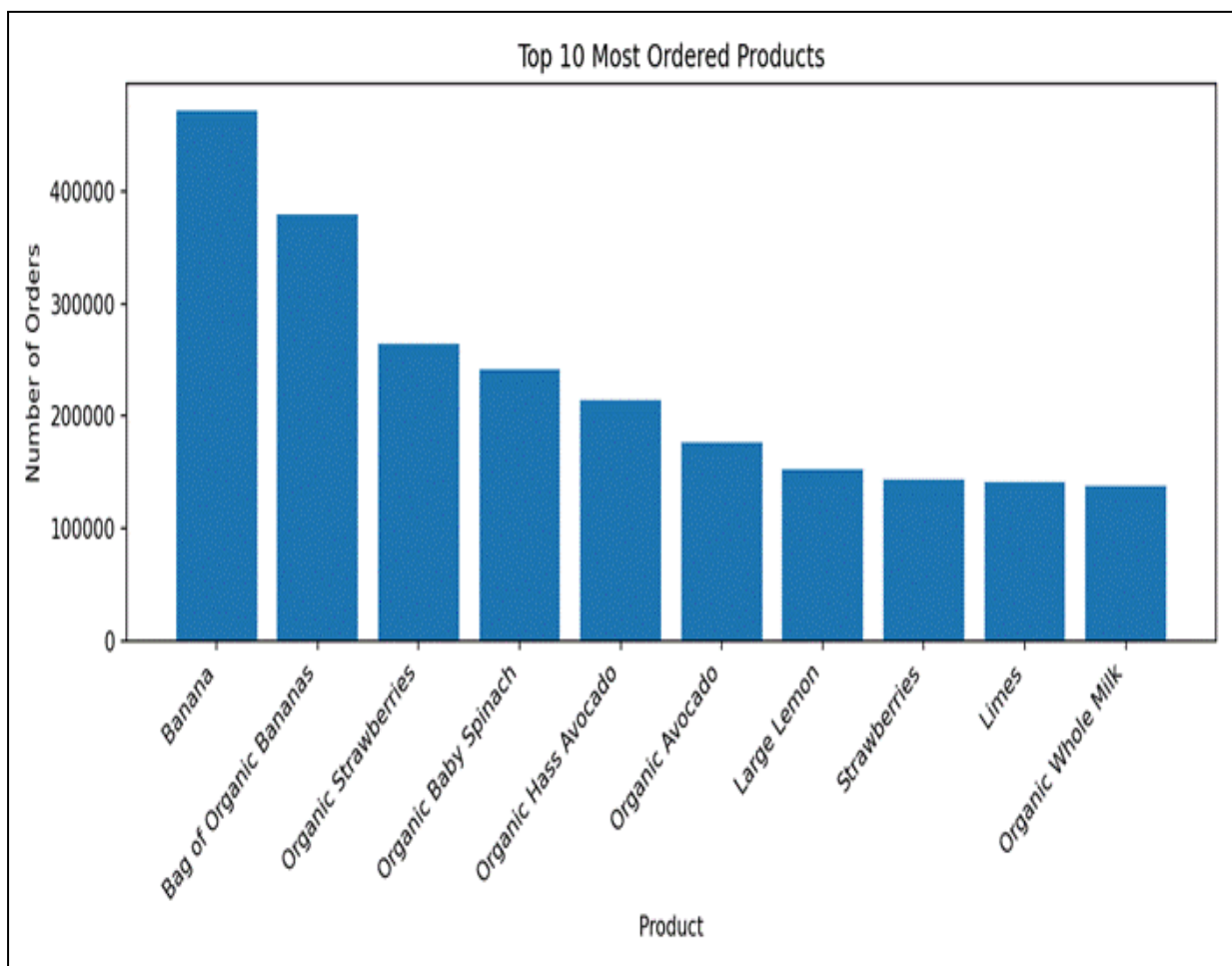


Figure 2 : Top 10 Most Ordered Products

6. Purchase Behaviour Analysis:

----- PURCHASE BEHAVIOUR -----	
New Purchases	13307953
Reordered	19126536

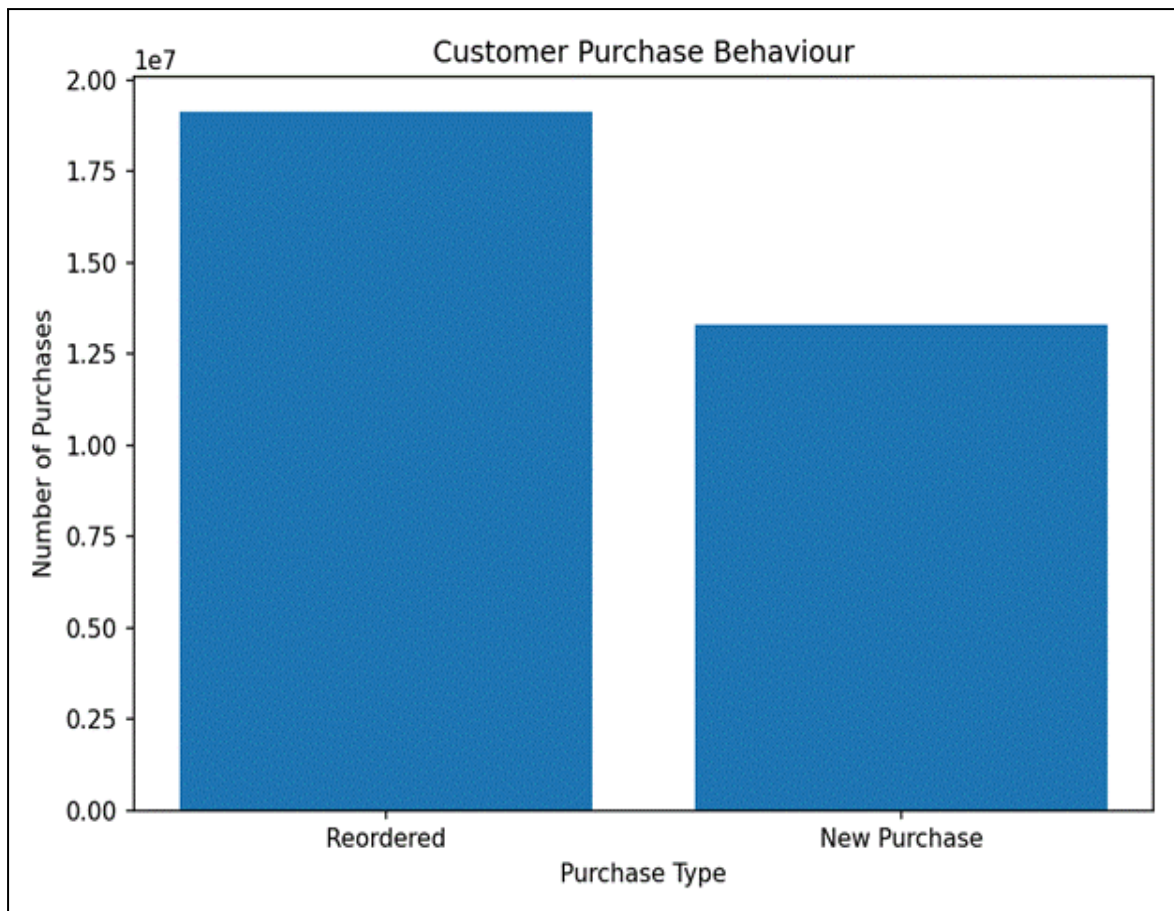


Figure 3 : New Purchases vs Reordered Items

----- DEPARTMENT ANALYSIS -----

```
department
personal care    6563
snacks           6264
pantry           5371
beverages        4365
frozen           4007
dairy eggs       3449
household        3085
canned goods     2092
dry goods pasta  1858
produce           1684
Name: count, dtype: int64
```

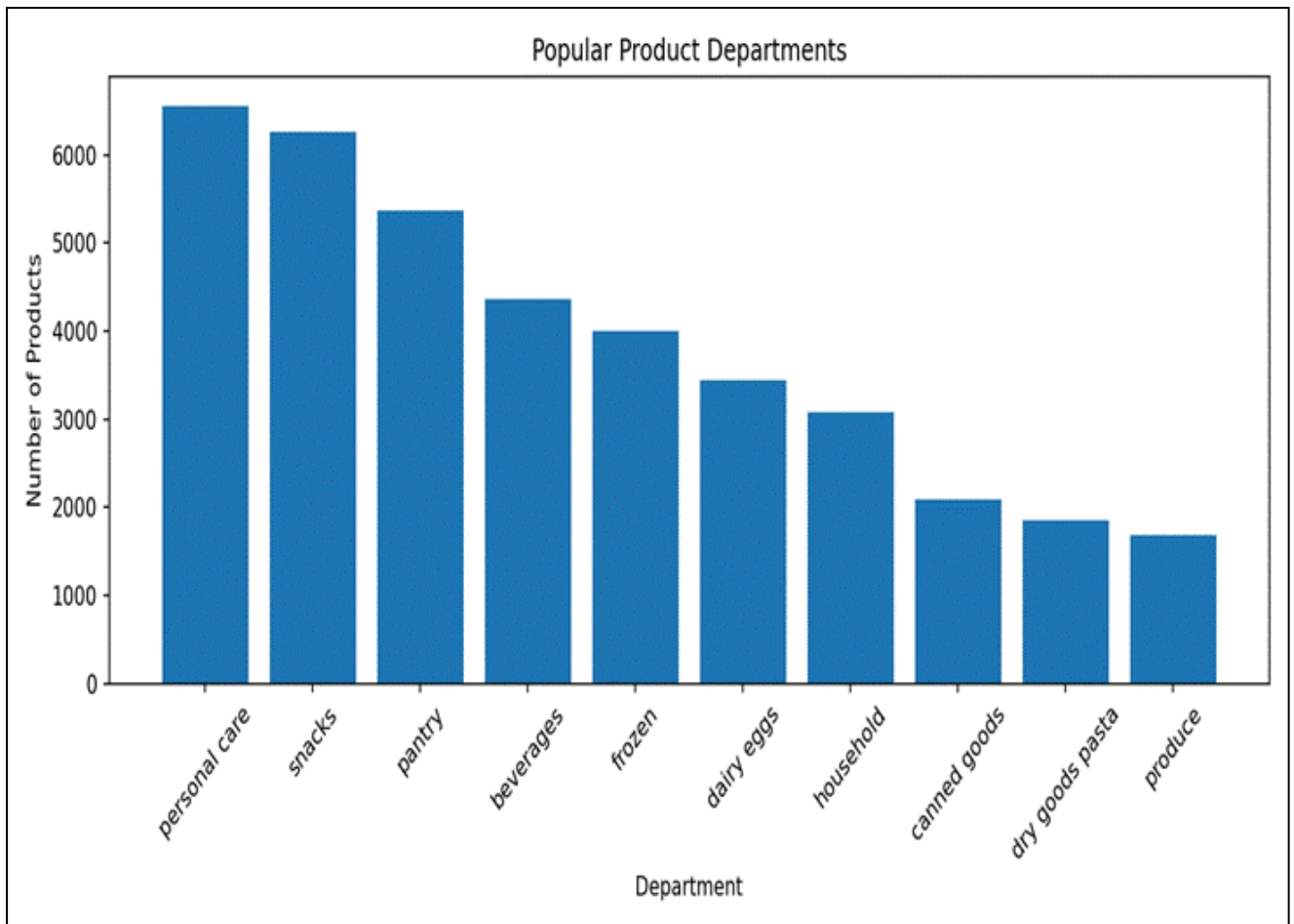


Figure 4 : Top 10 Product Departments

7. Trend Chart:

```
----- ORDERS BY DAY -----  
day_name  
Sunday      600905  
Monday      587478  
Tuesday     467260  
Wednesday   436972  
Thursday    426339  
Friday      453368  
Saturday    448761  
Name: count, dtype: int64
```

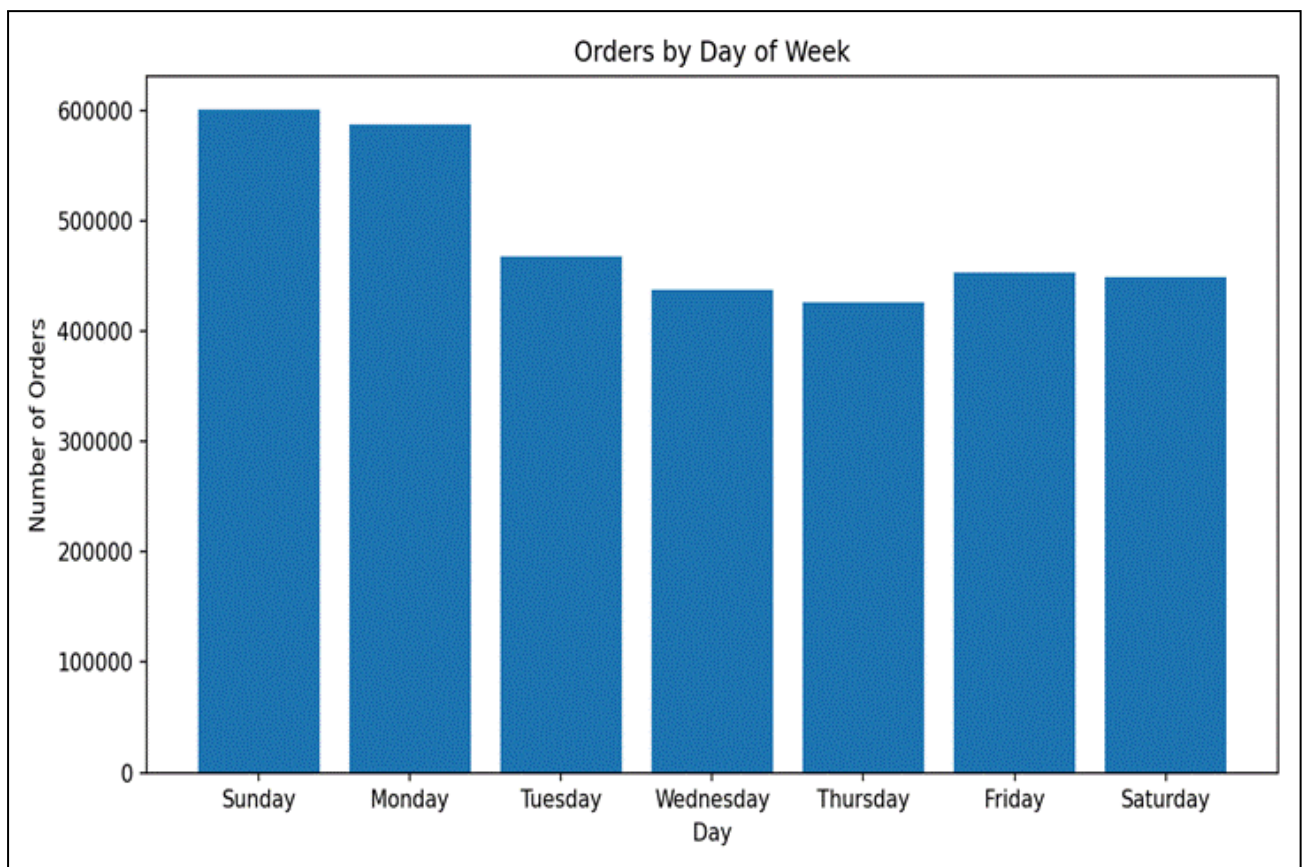


Figure 5 : Orders by Day of Week

```
----- ORDERS BY HOUR -----  
order_hour_of_day  
10      288418  
11      284728  
14      283042  
15      278073  
13      277663  
...  
Name: count, dtype: int64
```

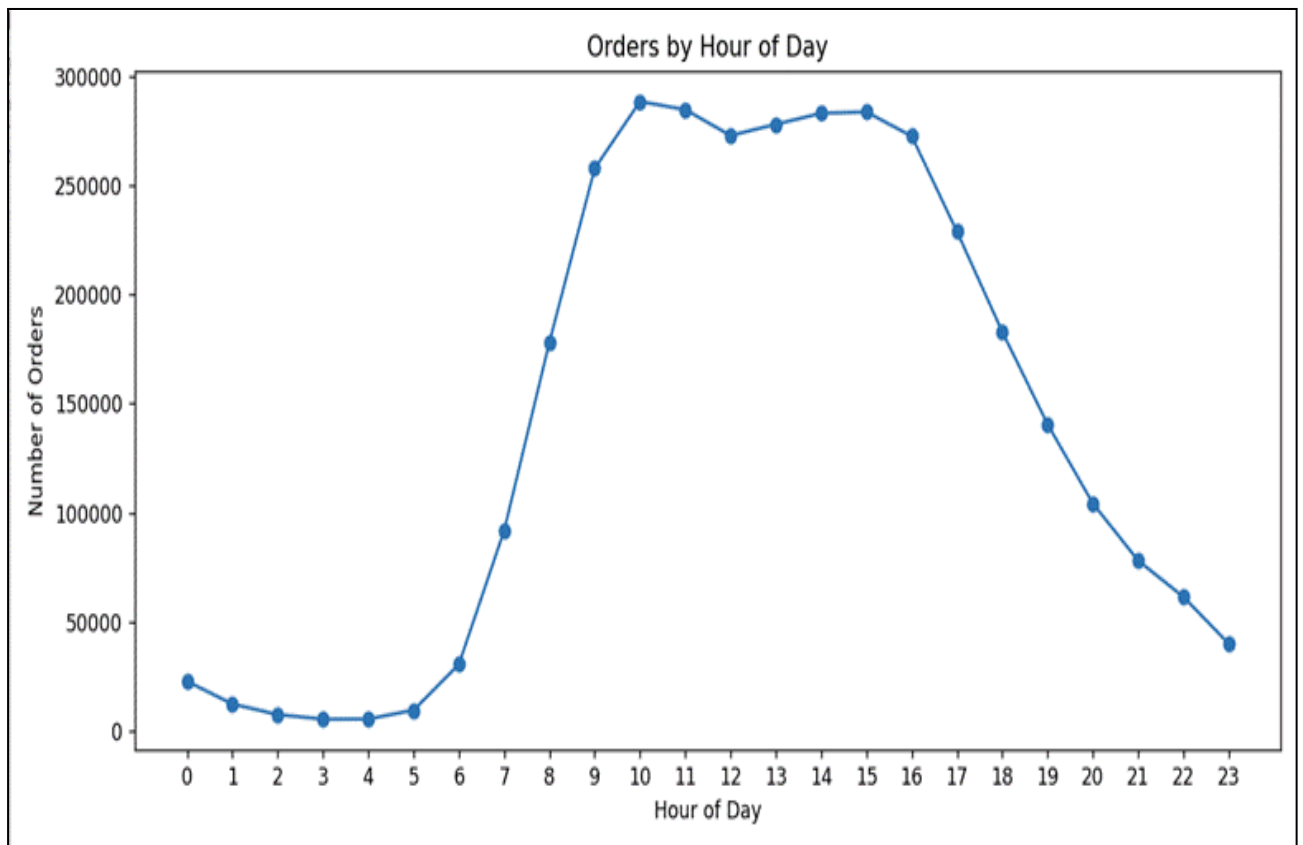


Figure 6 : Orders by Hour of Day

8. Order Analysis:

----- ORDER PATTERN -----

order_number

1 206209

2 156015

3 113757

4 87438

5 69263

...

Name: count, dtype: int64

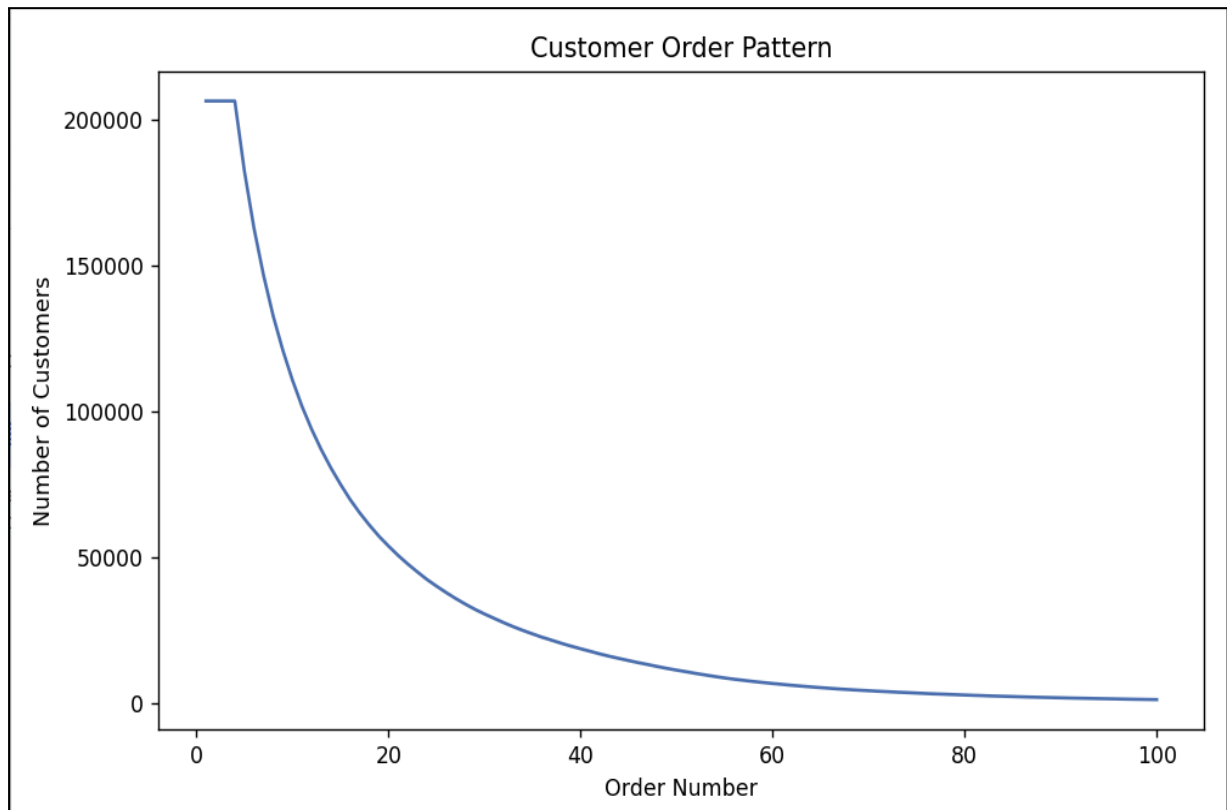


Figure 7 : Customer Order Pattern

9. Heatmap:

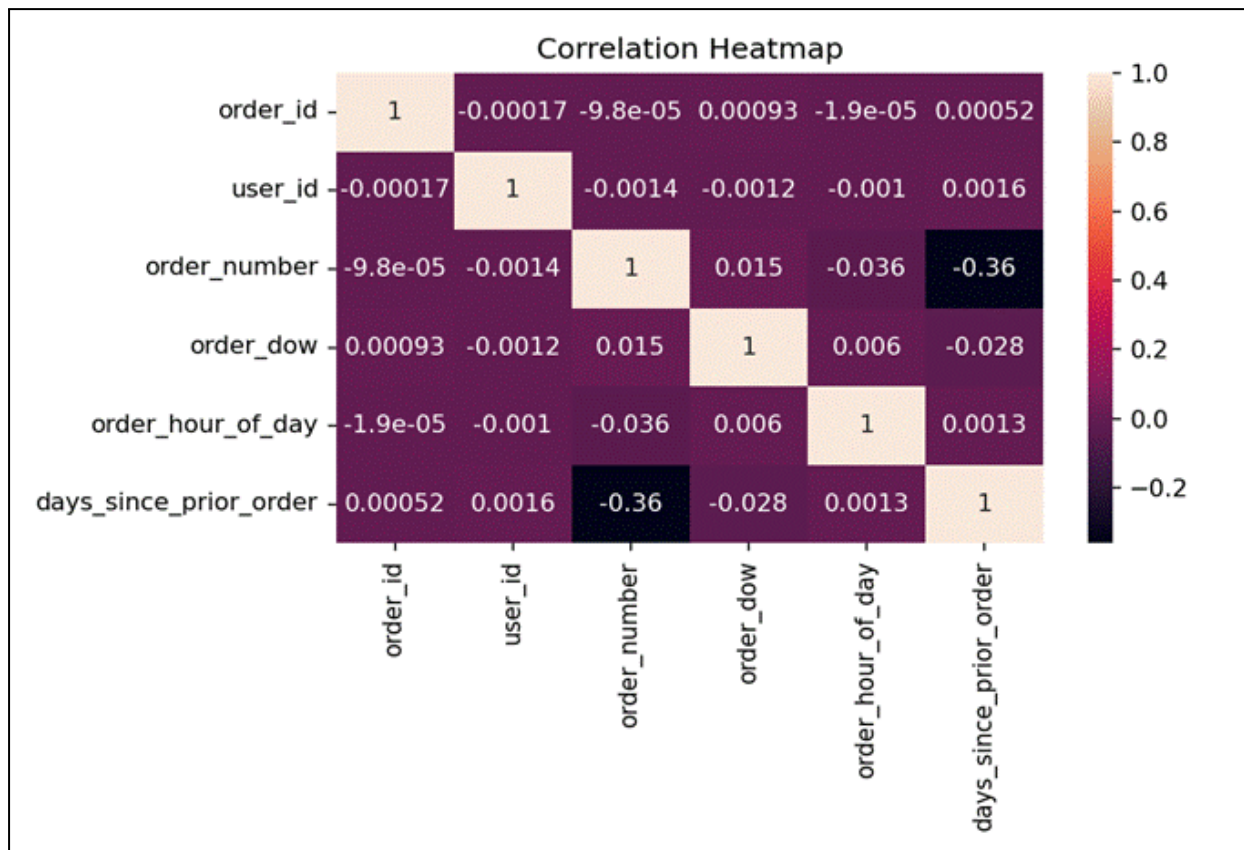


Figure 8 : Correlation Heatmap

10. Customer Indicators:

----- CUSTOMER INDICATORS -----

Total Orders 3421083
Total Customers 206209
Reorder Percentage 58.97%

11. Summary:

This project analyzes Instacart customer purchase behaviour using Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn to identify key shopping patterns and trends.

The analysis reveals that the dataset includes over **206,000 customers**, **3.4 million orders**, and **49,000+ products**, with an average of **16.59 orders per customer**, indicating strong customer retention. A significant **58.97% of purchases are reorders**, showing high product loyalty.

Bananas dominate product demand, with **Banana** and **Organic Bananas** being the most frequently ordered items. Shopping activity peaks on **Sunday and Monday**, with most orders placed between **10 AM and 3 PM**. Departments such as **Personal Care** and **Snacks** show the highest product variety.

Overall, the analysis provides valuable insights into customer behaviour that can support better recommendation systems, inventory planning, and business decision-making.